Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/24/2007
PI Name:	Mulloth, Lila M M.S.		
Project Title:	Development of a Solid-State Compressor for Closed-Loop Air Revitalization Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	lmulloth@mail.arc.nasa.gov	**Fax:**	FY 650-604-1092
PI Organization Type:	NASA CENTER	**Phone:**	650-604-1904
Organization Name:	NASA Ames Research Center		
PI Address 1:	M/S 239-15		
PI Address 2:			
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	02/01/2003	**End Date:**	02/01/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	John E. Finn		
COI Name (Institution):	LeVan, Martin Ph.D. (Department of Chemical Engineering, Vanderbilt University) Knox, James (NASA Marshall Space Flight Center)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The temperature-swing adsorption compressor (TSAC) is a solid-state compressor that has the capability to remove CO2 from a low-pressure source and subsequently store, compress, and deliver at a higher pressure as required by a processor. The TSAC is an ideal interface device for CO2 removal and reduction units in the air revitalization loop of a spacecraft such for oxygen recovery. The concept of TSAC as a potential interface for the carbon dioxide removal and reduction units in a life support system has already been proven and its operation has been demonstrated The ultimate objective of the task is to build an Engineering Development Unit (EDU) of an air-cooled TSAC with flight-like components and configuration and to test it in a relevant environment for certifying the technology readiness level of 6 (TRL 6) for this device. This two-bed prototype of TSAC will have a configuration that permits continuous		

	acquisition of CO2 from a low-pressure source such as CDRA and continuous supply of CO2 to the Sabatier reactor at the specified pressure, and will be sized to support a 4-person crew load in the ISS. The project involves three phases such as thermal model development and validation, full prototype design and testing, and EDU development This technology is important to closure of the oxygen loop on International Space Station which may offer substantial cost savings due to reduced logistics. Impact on future missions is likely to be higher
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Technology has application in energy (hydrogen) storage, heat pumps, sorption coolers etc.
Task Progress:	The objective of the project was to develop an air-cooled temperature-swing adsorption compressor (TSAC) that has application in closing the air revitalization loop in a spacecraft. Three phases were planned for this project with the following targets. Phase I: Development of the thermal model, mechanical design, and model validation (testing) of the air-cooled TSAC Phase II: Fabrication and testing of a full-size prototype to process 4kg CO2 per day Phase III: Develop an engineering development unit (EDU) with a technology readiness level (TRL) 6. Phase I targets were met on schedule on December 15, 2003. Test results indicated that efficiency of the compressor exceeded the design specification by about 50% with less than 30% of the estimated power. Results of the thermal model validation experiments were presented at the Habitation 2004 Conference at Orlando, Florida on January 5 and at the ALS workshop at Gainesville, Florida on January 9. Under phase II, a full-size (sized to serve the International Space Station at a full-crew load) prototype was built and tested in 2004. Results of full-size system design and initial test data were presented at the International conference on Environmental Systems, Colorado Springs, Colorado on July 20, 2004. The validation tests on the compressor were completed successfully on December 31, 2004. After the validation tests, the compressor was reassembled in a shipping crate to prepare for a relevant environment testing at the Space Station Simulation Laboratory at NASA Marshall Space Flight Center. The relevant environment tests on the TSAC were completed successfully in February 2006. The tests were performed in two phases. The Phase I tests were completed in December 2005 and phase II tests were completed in January 2006. The test results were presented in the International conference on Environmental Systems, Norfolk, Virginia on July 14, 2006. Phase I tests included the TSAC performance characterization in integration with the flight-like CDRA. Phase II experiments included integration tests of the TSAC with the CDRA and the Sabatier reactor EDU (Engineering Development Unit). The integrated tests of the TSAC with the CDRA and the Sabatier EDU were the precursor for the third phase of the TSAC EDU development process. The EDU development phase is on hold due to lack of or shortage of funds.
Bibliography Type:	Description: (Last Updated: 04/24/2007)
Papers from Meeting Proceedings	Mulloth LM, Rosen M, Varghese M, Luna B, Knox JC. "Performance Characterization of a Temperature-Swing Adsorption Compressor for Closed-Loop Air Revitalization Based on Integrated Tests with Carbon Dioxide Removal and Reduction Assemblies. " Presented at the 36th International Conference on Environmental Systems (ICES), Norfolk, Virginia, July 2006. SAE paper 2006-01-2126, July 2006. , Jul-2006
Papers from Meeting Proceedings	Knox JC, Miller L., Mulloth LM, Luna B, Campbell M. "Integrated Test and Evaluation of a 4-Bed Molecular Sieve (4BMS) Carbon Dioxide Removal System (CDRA), Temperature Swing Adsorption Compressor (TSAC), and Sabatier Engineering Development Unit (EDU)." 36th International Conference on Environmental Systems (ICES), Norfolk, Virginia, July 2006. SAE paper 2006-01-2271, July 2006. , Jul-2006